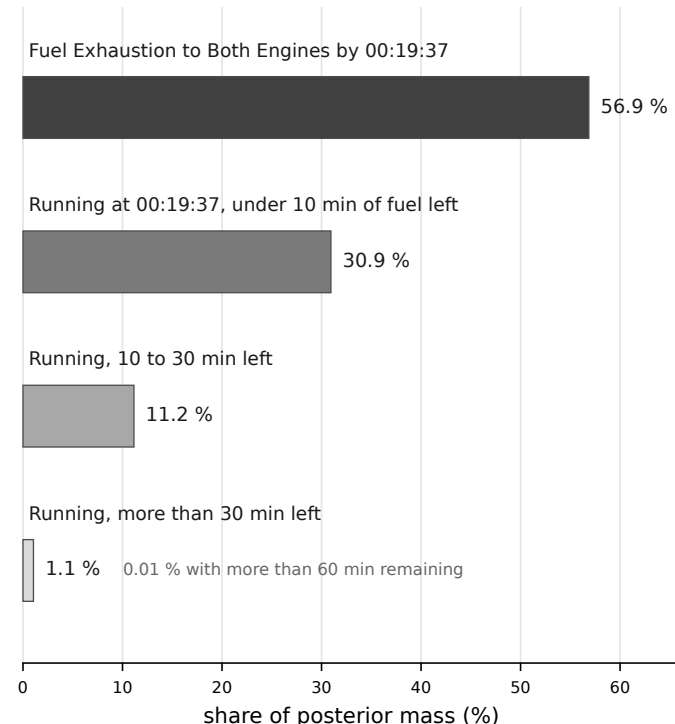
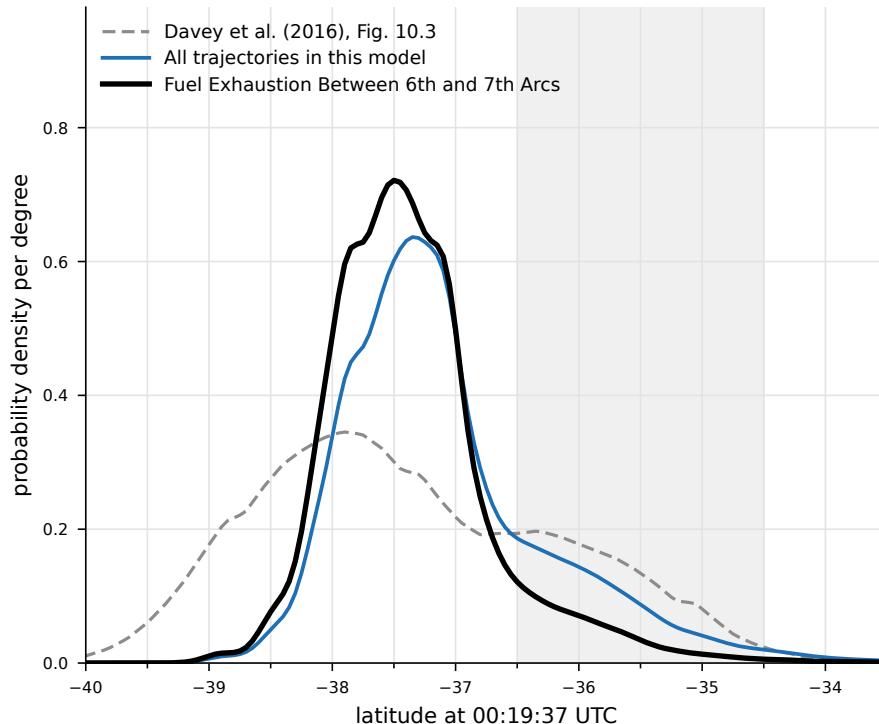




Fuel exhaustion (FE) is an OUTPUT of this model, not an input. The 00:17:30 exhaustion-time term is absent, so the share of trajectories reaching FE between the 6th and 7th arcs is a result rather than a condition imposed on the ensemble. Selecting that subset is therefore a post hoc selection.

FE before 00:11:00 is excluded by the data rather than assumed away: the aircraft transmitted at 00:11, so any trajectory already dry carries a log-likelihood penalty of -50 , a weight of about 2×10^{-22} . The residual mass with FE earlier than 00:09:36 is 0.46 %.

FE time is stored at 128 s resolution, so only about 21 times are representable. Of the 56.9 % reaching FE by 00:19:37, 51.0 % falls in slots wholly inside 00:11:00–00:19:29 and 5.4 % in the single slot spanning 00:09:36–00:11:44, which straddles the 6th arc. That 5.4 % is a storage artefact, not a distinct category. FE time over all trajectories — recorded where FE occurs by 00:19:37, projected from each trajectory's own mean burn rate where it does not — has median 00:19:12, mean 00:20:13 and a 95 % interval of 00:10:40 to 00:41:36; the lower quantiles inherit the same 128 s granularity.

Left panel: median latitude -37.23° over all trajectories and -37.43° over those reaching FE between the arcs, with 50 % intervals of 0.85° and 0.75° . Right panel: 98.9 % of the posterior is within 30 minutes of FE at the time of the final transmission, and 87.8 % within 10 minutes.